Unit: 6 Unit 2: Exponents Radicals
Topic: Operations with Radicals

Name: ________________
Date: ____________

Homework (Jalapeno)

Q1. Simplify $\sqrt{16} + \sqrt{9}$
(A) 7
(B) 5
(C) 4
(D) 3
(E) 1

Q2. The ancient Babylonians used $\sqrt{64}$ to represent the area of a square with side length 8. What is $\sqrt{64} \cdot \sqrt{25}$?
(A) 40
(B) 50
(C) 60
(D) 70
(E) 80

Q3. Simplify $\frac{\sqrt{36}}{\sqrt{4}}$
(A) 3
(B) 2
(C) 1
(D) 6
(E) 9

Q4. The ancient Egyptians used $\sqrt{49}$ to calculate the area of a pyramid's base. What is $\sqrt{49} - \sqrt{9}$?
(A) 2
(B) 4
(C) 6
(D) 8
(E) 10

Q5. Simplify $\sqrt{25} \cdot \sqrt{9}$
(A) 15
(B) 20
(C) 25
(D) 30
(E) 45

Q6. What is $\sqrt{16} \div \sqrt{4}$?
(A) 2
(B) 4
(C) 6
(D) 8
(E) 10

Q7. The ancient Greeks used $\sqrt{81}$ to calculate the area of a circular statue base. What is $\sqrt{81} - \sqrt{16}$?
(A) 5
(B) 7
(C) 9
(D) 11
(E) 13

Q8. Simplify $\sqrt{100} + \sqrt{1}$
(A) 10
(B) 11
(C) 12
(D) 13
(E) 14

Open Ended Questions (FRQ)

Q9. The ancient Romans built an aqueduct with an arch that had a height of $\sqrt{144}$ meters. If the width of the arch was $\sqrt{49}$ meters, what was the area of the arch?

Q10. A pharaoh's tomb has a square base with an area of $\sqrt{256}$ square meters. If the length of one side of the base is $\sqrt{16}$ meters, what is the length of the other side?

Chef's Tips

- Tip 1: Encourage students to simplify radicals before performing operations.
- Tip 2: Emphasize the importance of using the properties of radicals to simplify expressions.
- Tip 3: Remind students to check their work and ensure that their answers are in simplest radical form.